
MNL-66(20) / ACI 318-19 STEP-BY-STEP DETAILING

Reinforcement t Verification Manual

A step-by-step walk of every reinforced-concrete element in WebSketch 3D - beam, column, wall, slab, foundations and their floor-to-floor connections - with app renders, the exact code path for each step, and a verify checklist per detail.

WebSketch 3D - 5-story reference building -
September 2026

WEB-SKETCH-3D - DOC RV-MNL66-001

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1. How to Use This Manual

This manual is a verification aid: it walks the reinforcement of each element step by step, in the order the generator builds it, with a picture of the real app output and the exact code that produces each step. For every detail there is a **VERIFY** box - the detail is correct if the listed conditions hold on screen. The reference scene is **File > Load MNL-66 Reinforcement Demo** (a 5-story building, every feature present).

1.1 Golden rules of the app

- **Framing trim:** a column on the beam line divides the beam - the beam **SOLID** ends at the column face; the analytical baseline keeps running center-to-center. `StructuralManager.js > buildBeam > colReach()`
- **Columns run through:** the column occupies the full story; beams, slabs and walls connect **TO** it.
- **Rebar display:** **Light** = centerline ribbons (fast), **Detailed** = solid pipes. Toggle: **View > Rebar: Light / Detailed**.
- **X-Ray** reveals interiors; in Detailed mode rebar renders as an opaque red pass over the ghosted concrete.
- **Units:** meters on screen and in code; detailing rules quoted from ACI 318-19 and MNL-66(20) (inches in the book, converted here).

1.2 Verification workflow

1. Load the demo (about 18 s; opens in X-Ray + Light rebar).
2. Zoom to the element under verification (chapter coordinates given per figure).
3. Switch to Detailed rebar when checking a single bar closely.
4. Read the chapter steps top to bottom - they follow the code path exactly.
5. Tick the **VERIFY** boxes. Anything that fails is a bug: note the element, the step number, and the figure.

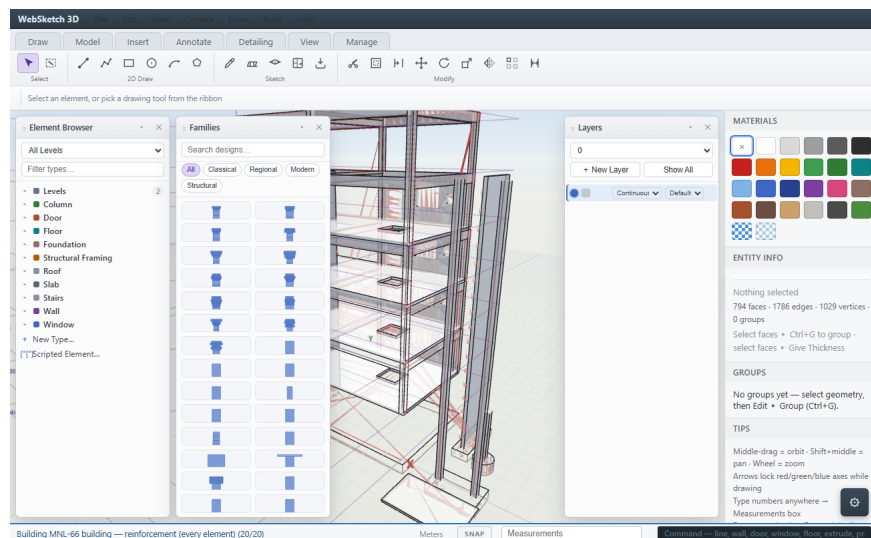


Figure 1-1 - The 5-story reference building (Light rebar, X-Ray).

2. Beam (start here)

Rules: ACI 318-19 Ch. 9 + 25; MNL-66(20) BM-202/203/204, Table 25.3.1

The beam generator is `buildBeamRebar()` in `js/features/elementrebar.js`. It receives the beam entity, reads the framing-trimmed solid, and builds the cage in this order:

2.1 Step by step

1. **Find the end section.** `endFaceOfBeam()` picks the cap face perpendicular to the run; `faceFrame()` gives section coordinates (u across, v vertical). For T/L beams the frame widens to the UNION of all cap fragments, so u spans the full flange.
2. **Rows and covers.** Bottom row sits at $\text{cover} + \text{tie} + \text{db}/2$ above the soffit; top row the same below the top face. T/L: the tie cage wraps the WEB (`webWidth`), the top row spreads the FLANGE.
3. **Stirrups - one closed loop.** `stirrupPath()` (`js/features/rebar.js`): all four corners are nesting arcs - the bend center coincides with the corner bar axis; 135 degree lap hooks at alternating top corners; hook extension $\max(6\text{db}, 75 \text{ mm})$.
4. **Stirrup spacing.** Non-seismic: user spacing/amount via `distribute()`. Seismic toggle: first tie 50 mm off the support face, confinement zones $2h$ at $\min(d/4, 125 \text{ mm})$, mid-span at $d/2$ (`seismicTiePositions()`).
5. **Multi-leg rule (300 mm).** If the clear transverse distance between outer tie legs exceeds 300 mm, inner hoops are generated so no leg spacing exceeds it.
6. **Longitudinal bars.** Continuous bars the full span; extras curtailed at $L_n/4$ (top, at supports) and $L_n/8$ (bottom, off supports); second layer with 25 mm clear when the row is too crowded.
7. **End hooks (Table 25.3.1).** 90 deg: tail 12db vertically into the core; 180 deg: return leg toward the core. Bend centerline radius 3.5db (6db inside diameter).
8. **Connection - BM-202 far side.** The generator looks for the column at each end (entity pool, 0.75 m probe). At a supported end BOTH rows run on to the far side of the column ties ($\text{center} + \text{plan reach} - 40 \text{ mm cover} - \text{tie dia}$) and take the standard 90 deg hook there.
9. **Connection - BM-204 girder ends.** A perpendicular beam crossing at the end receives the bars the same way (half of its `webWidth` less covers).
10. **Tie cap (BM-203/204).** Non-seismic: through each supported 0.6 m connection zone, tie spacing tightens to 8 in (203 mm) max.
11. **Integrity steel (ACI 9.8.1.2).** At least 2+2 continuous bars; at a free (unsupported) end both rows get 90 deg hooks.
12. **Skin steel (ACI 9.7.2.3).** Auto when $d > 0.9 \text{ m}$: side bars over $h/2$ at max 0.25 m.
13. **Cranked bars.** Optional 45 deg bent-up bars, hooks down at the top elevation.

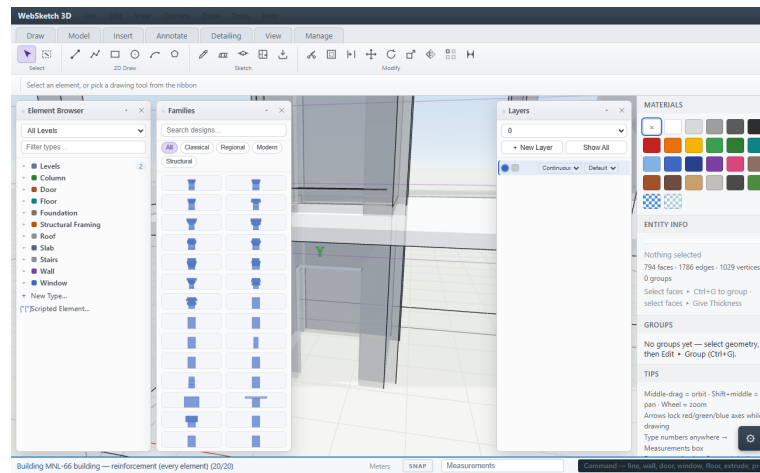


Figure 2-1 - Beam cage: closed stirrups, alternating lap corners, top/bottom rows at the covers.

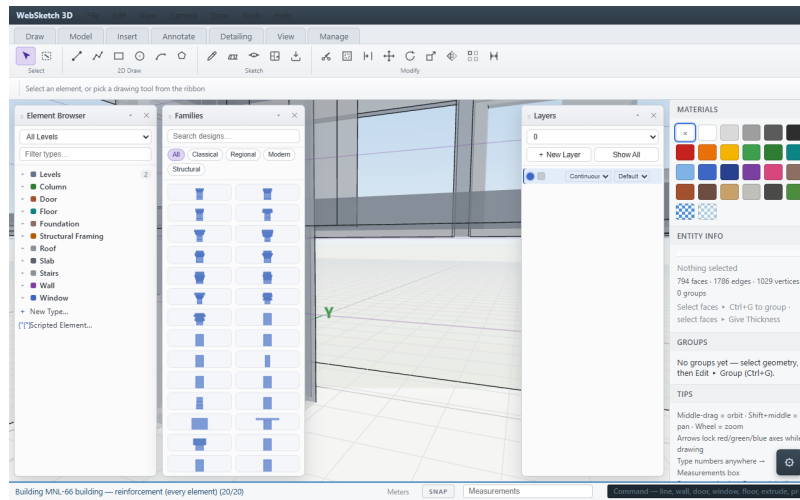


Figure 2-2 - BM-202: bars pass the column face and hook at the far side of the column ties.

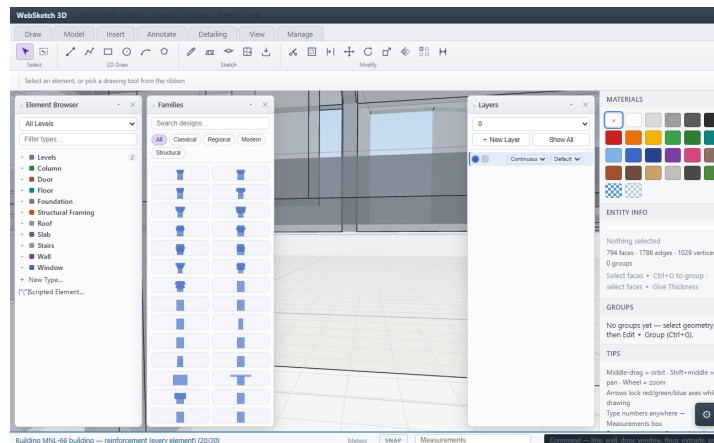


Figure 2-3 - End hook: 12db tail, bend on the corner-bar axis.

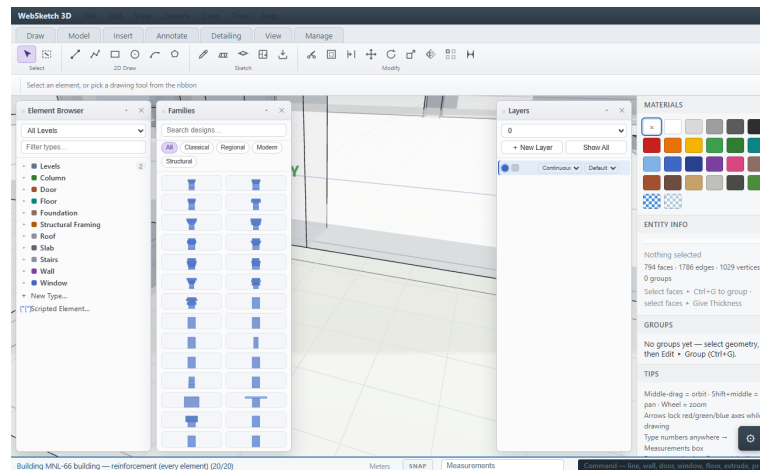


Figure 2-4 - T-beam: cage wraps the web, top row spreads the flange.

VERIFY - the detail is correct if:

- Stirrup corners wrap the corner bars (bend center ON the bar axis), not floating inside.
- 135 degree lap hooks alternate top corners between consecutive stirrups.
- Every bar entering a column CONTINUES past the solid face and hooks inside the column (no bar stops at the face).
- Bars never cross the concrete surface anywhere except at their hooked tips.
- At the supported 0.6 m zones, stirrup spacing visibly tightens to 8 in or less.
- Free beam ends still show 90 deg hooks (integrity steel).

3. Column

Rules: ACI 318-19 Ch. 10 + 25.5; MNL-66(20) COL-100/200/202

Column cages come from `buildColumnCage()` in `js/features/columnrebar.js` (4 cage types: single tie, two ties, multiple sets, circular helix). The element tool wraps it via `buildColumnRebar()`.

3.1 Step by step

1. **Section framing.** Pick the top face; frame from it. A genuinely circular section (8+ ring points at constant radius - `isCircularLoop()`) switches to the helix cage automatically.
2. **Ties.** `stirrupPath()` on the column section - one closed loop, 135 degree laps; ties distribute along the height by spacing or amount.
3. **Corner + inner bars.** 4 corners always; additional sets on each face for the "multiple" type. Corners are never lighter than inner bars (corner dia = max dia in cage).
4. **Splice (COL-200).** Lap mode: each bar splits into two pieces just above the floor; lap length = $1.3 \times 47.5 \times \text{psi-s} \times \text{db}$ (Class B, Grade 60, 4 ksi; $\text{psi-s} = 0.8$ up to 19 mm), min 300 mm. Alternate bars stagger by half a lap; 2 extra ties through the splice zone; short columns keep whole bars.
5. **Splice (COL-202).** Mechanical / end-bearing: one square-cut bar + a 150 mm sleeve at the splice plane, no stagger.
6. **Base of column.** If a foundation sits directly below, the base lap is DROPPED - the footing starters are the lower splice piece (`buildColumnRebar()` detects the foundation). Bars run whole and lap the starters.

7. Story columns. One entity per story: every column above floor 1 splices onto the slab (no foundation below) - this is the floor-to-floor connection you see repeating at each level of the demo.

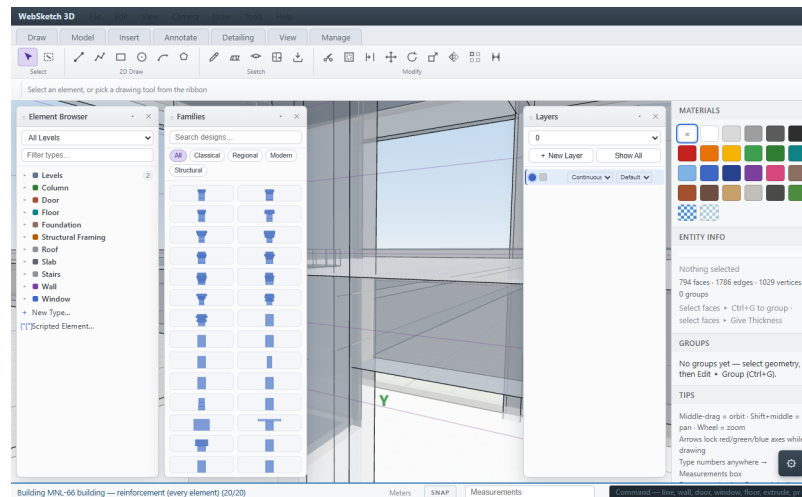


Figure 3-1 - Story splice above the slab: staggered laps + the two extra splice-zone ties.

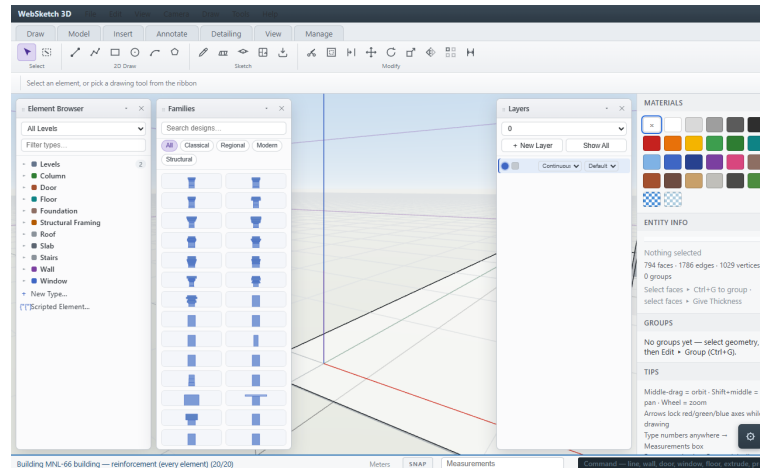


Figure 3-2 - Column base on a foundation: whole bars lapping the L-starters (no duplicate base lap).

VERIFY - the detail is correct if:

- Lap pieces overlap the FULL lap length (0.79 m for 16 mm bars) - measure between piece tips on the same line.
- Two splice heights alternate around the section (stagger = half lap).
- Two extra ties sit in the lowest 0.4 m of the splice zone.
- Over a footing: starters + whole bars, never three bars stacked on one line.
- Helix (if any circular column): constant pitch, 1.5 extra turns top and bottom.

4. Wall

Rules: ACI 318-19 Ch. 11; MNL-66(20) WALL-100A/110/201/206/207/208

Wall rebar is `buildWallRebar()` in `elementrebar.js`. Two curtains (each face) by default.

4.1 Step by step

1. Minimum ratios. Vertical $\rho \geq 0.0012 \times t$, horizontal $\rho \geq 0.0020 \times t$; spacing caps at $\min(3t, 450 \text{ mm})$. The generator TIGHTENS spacing to satisfy the ratio before placing bars.

2. **Boundary elements (WALL-110).** Shear walls (boundary toggle) add 2 concentrated verticals inside each end, each curtain.

3. **Openings (WALL-206/207/208).** Hosted door/window/opening entities on the wall are read directly: the mesh SPLITS around each hole (150 mm / 250 mm sliver rules); 2 horizontal bars at head and sill + 2 vertical bars each jamb, developing 24 in past the opening; a 48 in diagonal per corner per curtain at 45 degrees (clamped to fit, 24 in minimum); floor-level openings get no sill steel.

4. **Circular openings (WALL-201).** shape:"circle" openings trim around the disc the same way (bars each side of the pipe).

5. **Starter dowels (WALL-100A).** A slab or foundation whose top sits at the wall base gives one L-dowel per vertical station per curtain: leg inside the host, rising the Class B lap into the wall.

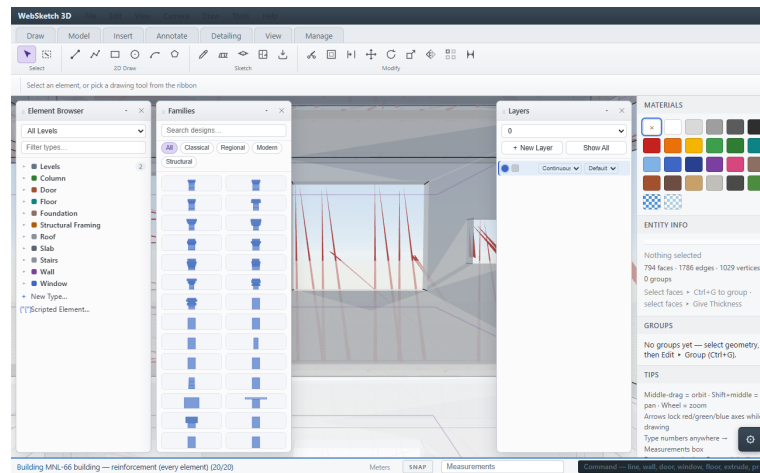


Figure 4-1 - Facade bay per floor: door, window and circular pipe opening with trim pairs + diagonals.

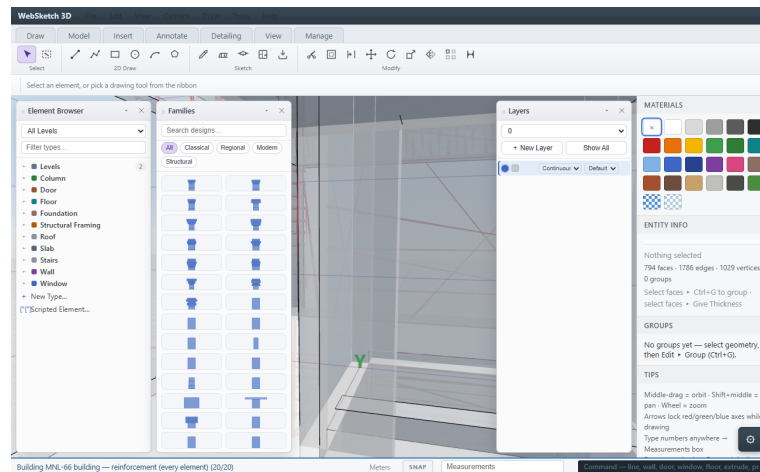


Figure 4-2 - Elevator shaft corner: boundary elements concentrated at the wall end.

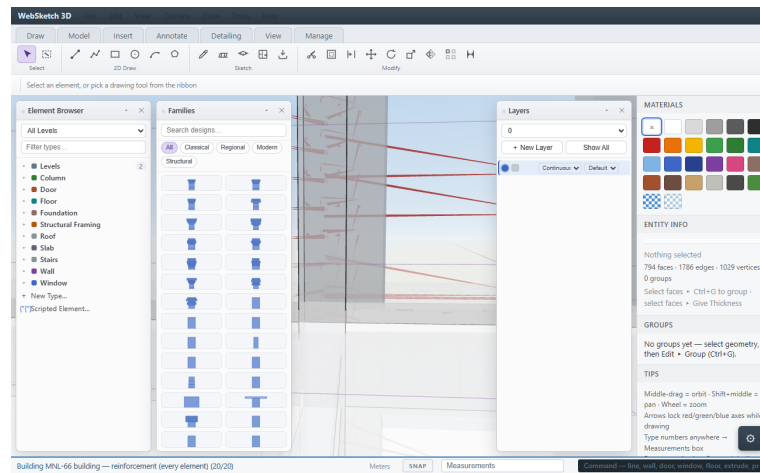


Figure 4-3 - WALL-100A: wall starter dowels hook into the slab and lap the wall verticals.

VERIFY - the detail is correct if:

- No mesh bar crosses an opening; verticals resume above/below, horizontals run left/right.
- Trim pairs (2 bars) appear at head/sill/jambes, extending past the opening.
- 45 degree diagonals at opening corners, never poking outside the wall.
- Wall-over-slab joints show L-dowels crossing the joint plane.
- Shear wall ends show denser vertical concentration (boundary bars).

5. Slab / Floor

Rules: ACI 318-19 Ch. 7/8 ($A_s \geq 0.0018 A_g$, $s \leq \min(3h, 450 \text{ mm})$); MNL-66(20) SLAB-200/206

1. **Mesh.** Two-way bottom mesh clipped to the slab outline and around every hole (scanline even-odd clipping); Y layer on the cover, X layer resting on it; top mesh optional.
2. **Openings (SLAB-206).** Bars split at holes; trim bars (one per side + side bars at pitch) span the hole + development each side; slivers under 250 mm drop.
3. **Corner reinforcement (SLAB-200/201, ACI 8.6.1.2).** Top and bottom bars extending the long span/5 from every exterior corner, both directions.
4. **Circular holes.** Clipped like any hole (via the entity sketch when the curved ring loses remaps); trim bars ring the disc.
5. **Slab-on-ground (SOG-102/103/105).** Toggle: mesh stays CONTINUOUS through the thickened edge; 2 continuous edge bars per edge lapped $\min(0.6 \text{ m}, L/4)$ past corners, clipped inside.
6. **Column punches.** Slabs are punched at every column with a 3 mm construction reveal (keeps the holed top face alive); the mesh respects the punches.

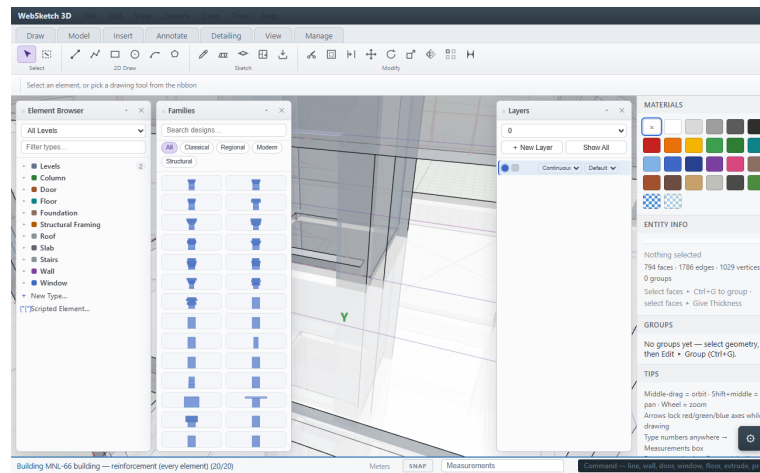


Figure 5-1 - Two-way mesh clipped to the outline and the openings.

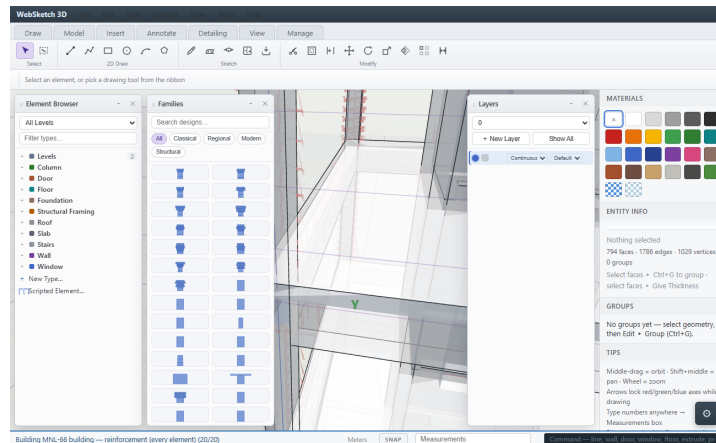


Figure 5-2 - Corner reinforcement at an exterior corner.

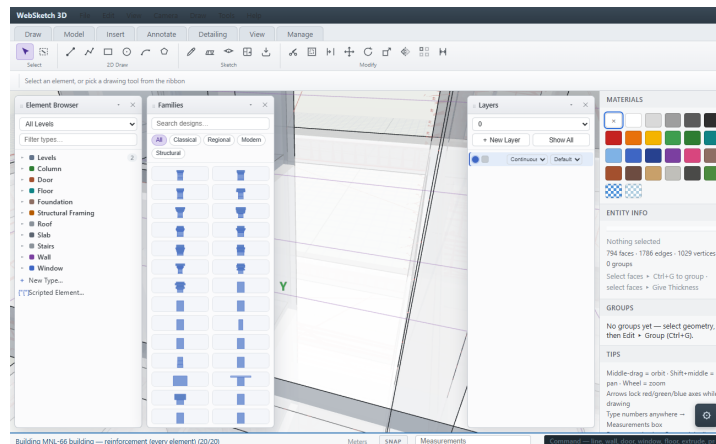


Figure 5-3 - Circular penetration: mesh breaks, trim bars ring the disc.

VERIFY - the detail is correct if:

- No bar crosses a hole of any shape; rows split cleanly at the hole boundary.
- Corner zones show 4-way bars (X and Y, top and bottom) within the corner band.
- SOG apron: perimeter bars follow every edge including lapped corners.

6. Foundations

Rules: ACI 318-19 Ch. 13; MNL-66(20) FND-102/103/109/150/161

1. **Pad (default).** Two-way bottom mesh (spacing caps min(3h, 450 mm), ratio 0.0018 enforced); L-starters for the column above with a Class B lap (0.69 m for 14 mm).
2. **Mat (FND-109).** kind:"mat" - the mesh runs at BOTH faces (top mesh under the top cover).
3. **Pile cap (FND-161).** kind:"pilecap" - piles on a centered grid; per pile 4-6 ring dowels lapped down into the shaft + circular shaft ties continued through the cap depth.
4. **Drilled pier (FND-150).** A CIRCULAR pad auto-detects: shaft verticals on a ring (max 200 mm spacing) + circular ties through the cap; the mesh clips to chords of the disc. With a column above, the shaft steel RISES the Class B lap above the cap so the column bars lap it.
5. **Strip footing (FND-102).** A wall above = a line of wall dowels along the run at the wall spacing.
6. **Combined footing (FND-103).** EVERY column on the pad gets its own starter group.

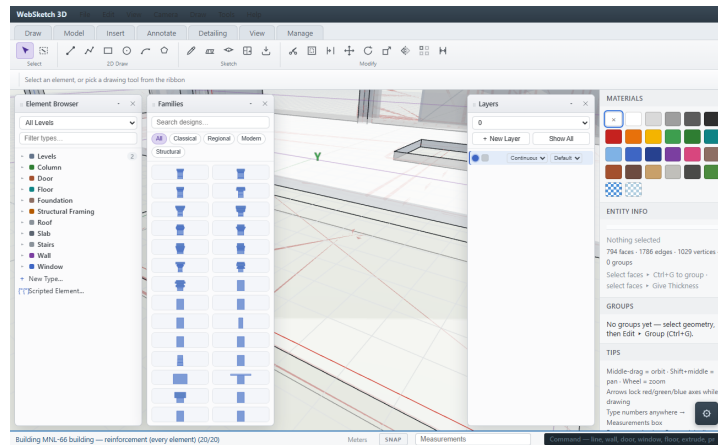


Figure 6-1 - Raft: mesh at both faces under the tower.

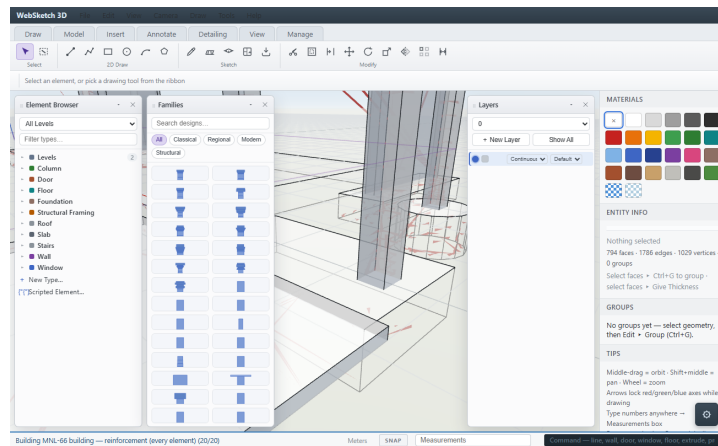


Figure 6-2 - Combined footing: two starter groups on one pad.

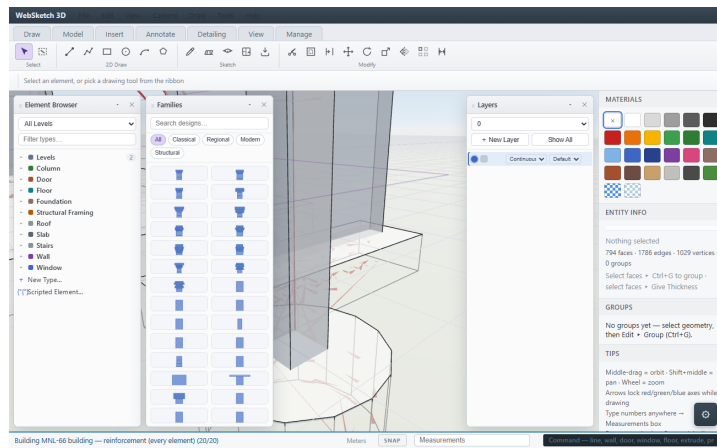


Figure 6-3 - Strip footing under the annex shear wall: dowels along the run.

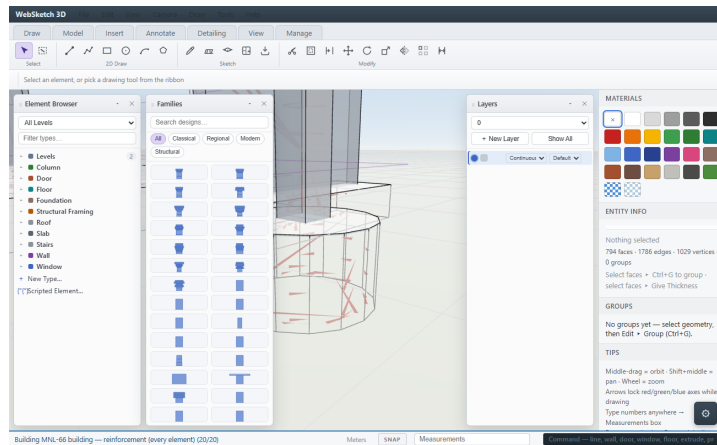


Figure 6-4 - Drilled pier: shaft ring + ties; shaft steel rises to lap the column.

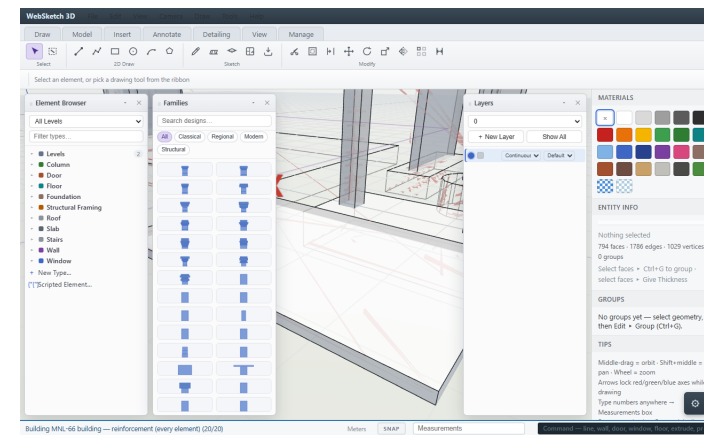


Figure 6-5 - SOG apron with continuous perimeter edge bars.

VERIFY - the detail is correct if:

- Starters rise the full Class B lap above the footing top (measure: ~0.69 m for 14 mm).
- Combined footing: starter groups under BOTH columns, at each column footprint.
- Pier: shaft verticals continue above the cap top where a column stands (no empty joint plane).
- Mat: identical mesh pattern visible just under the top face.

7. Floor-to-Floor Connections

Rules: BM-202/204, COL-200, WALL-100A - the "everything meets everything" chapter

The demo stacks five identical stories so every joint type repeats at $z = 3, 6, 9$ and 12 m. Walk one typical floor and check the steel crossing each joint plane:

- **Beam - column:** bars pass the face, hook at the far side of the column ties (Fig. 2-2).
- **Column - slab:** story column bars splice above the slab (Fig. 3-1); the slab is punched around the column.
- **Wall - slab:** L-dowels from the slab lap the wall verticals (Fig. 4-3).
- **Pier - column:** shaft steel rises through the cap into the column lap zone (Fig. 6-4).
- **Elevator lintel:** a real beam over every shaft door, bearing 0.25 m each side.

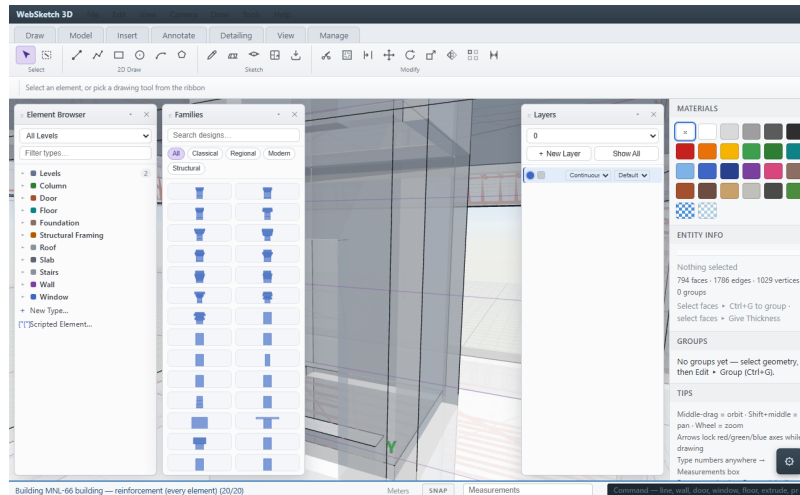


Figure 7-1 - Elevator shaft: shear walls, door openings, lintels, hoistway through the slabs.

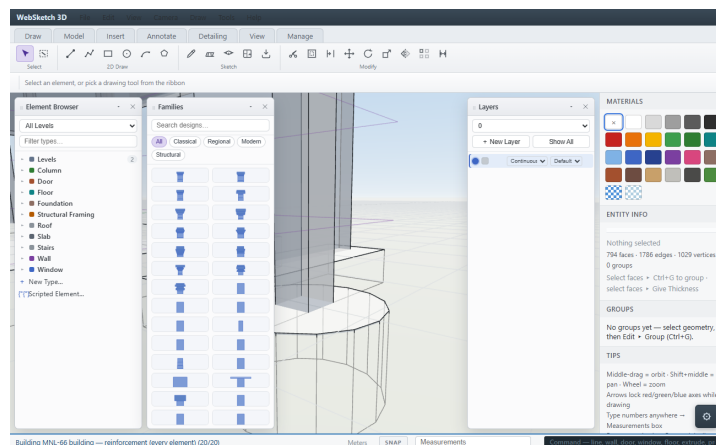


Figure 7-2 - The pier column splices onto the FND-150 shaft steel.

8. Schedules and the Bar Bending Schedule

Menu > Element Schedules (per ACI 315 4.10) and the BBS read the same rebar data as the drawings: every bar carries a pid; the BBS groups by host + shape + diameter + length, counts by pid (never by faces - one bar is many pipe faces). Record-only scenes (model.rebarPipes = false, used by the demo) feed the same tables from the bar registry. Cross-check: the count of a given bar in the schedule equals the pieces you count on screen.

9. Code Map

What	File	Function / site
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Beam cage, connections, integrity	js/features/elementrebar.js	buildBeamRebar(), endSupport(), addBar()
Column cage (4 types)	js/features/columnrebar.js	buildColumnCage(), verticalBar() (splices)
Wall rebar, openings, boundary, dowels	js/features/elementrebar.js	buildWallRebar()
Slab mesh, trims, corners, SOG	js/features/elementrebar.js	buildSlabRebar(), clipScanline()
Footings (all kinds)	js/features/elementrebar.js	buildFootingRebar()
Stirrup / bar shapes	js/features/rebar.js	stirrupPath(), roundedPath(), distribute()
Bar kernel (pipes / records)	js/model.js	addRebarPath()
Framing trim (beam ends at columns)	js/StructuralManager.js	buildBeam() colReach()
Hosted openings (doors/windows)	js/app.js	BimTools.HostedCut
Elevator shaft builder	js/features/elevator.js	ElevatorFeature.buildOne()
5-story demo scene	js/features/demo-mnl66.js	buildAsync()
Tests mirroring this manual	test/*.test.js	beamconn, columnrebar, walopening, story5, fndkinds, sog

Verification sources in the book: MNL-66(20) figures BM-202/203/204, COL-100/200/202, WALL-100A/110/201/206/207/208, SLAB-200/201/206, FND-102/103/109/150/161, SOG-102/103/105; ACI 318-19 Tables 25.3.1 and 25.5.5.1, sections 7.6.1.1, 8.6.1.2, 9.7.2.3, 9.8.1.2, 10.6.1, 11.6.1, 13.3.2. The parsed figure database lives in _mnl66_details.json (137 details).